ENGLISH TRANSLATION: JP S6165742 A

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Title: permanent magnet chuck

Claims
Claims (5)
Translated from Japanese
[Claims]
(1) A face plate having a plurality of magnetic bands extending parallel to each other with non-magnetic bands in between; a bottom plate made of a magnetic material disposed along the face plate; and a magnetic material between the bottom plate and the face plate; first and second magnetic yokes arranged alternately corresponding to the bands, one of the magnetic yokes being magnetically divided into an upper magnetic part and a lower magnetic part with a non-magnetic part in between; The yoke is disposed between each magnetic yoke and is movable in the vertical direction between the bottom plate and the face plate with the magnetic pole surface slidably in contact with the corresponding magnetic yoke, and in the direction in which the magnetic yokes are arranged. A permanent magnet chuck that includes a plurality of permanent magnets arranged with their magnetization directions alternately reversed, and an operation mechanism that moves the permanent magnets together in the vertical direction.
(2) The permanent magnet according to claim (1), wherein the height dimension of each of the permanent magnets is equal to or less than the height dimension of each of the upper magnetic part and the lower magnetic part of the magnetic yoke. Chuck.
(3) The first magnetic yoke is divided into the upper magnetic part and the lower magnetic part, and the upper end of the upper magnetic part returns to the corresponding magnetic band of the face plate, and A permanent magnet chuck according to claim 1, wherein the upper end is attributable to the corresponding magnetic zone of the face plate.
(4) The permanent magnet chuck according to claim (1), wherein each of the permanent magnets is fixedly supported by a frame that is integrally movable in the vertical direction between each of the magnetic yokes. .
(5) The permanent magnet chuck according to claim (1), wherein each of the permanent magnets is a high coercive force magnet.

Description
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Translated from Japanese
DETAILED DESCRIPTION OF THE INVENTION (Technical Field) The present invention relates to a hydraulic magnetic chuck that holds a magnetic material by magnetic force.
(Prior Art) In a conventional permanent magnet chuck that uses a permanent magnet assembly as a magnetic force source, for example, as disclosed in Japanese Utility Model Publication No. 10057/1980, there is a 1. A plurality of fixed permanent magnets arranged on the face plate so as to align the magnetic bands of the blood plate in a direction transverse to the width thereof, and a plurality of fixed permanent magnets arranged in parallel thereto in correspondence with the fixed permanent magnets; A movable magnet that can be integrally moved in a direction across the magnetic zone is provided. In the chuck, the movable permanent magnet is moved in a direction across the magnetic zone on a plane parallel to the face plate, thereby changing the relative position of the 1N Kicho Domizukyu magnet with respect to the fixed permanent magnet. The recording plate can be switched between an energized state and a non-energized state 5.

However, the i+J dynamic hydraulic magnet has a relationship between the face plate and γ Since the magnetic band located at the outermost part of one of the face plates is movable in the direction across the magnetic band of the face plate on the surface of the face plate, the fixed r The G stone is located at a position relatively protruding from the fixed magnet. Therefore, in the conventional chuck, the magnetic flux of the magnet cannot be effectively applied to the magnetic band located at the outermost part, and it is not possible to effectively magnetize all the magnetic bands of the face plate. .

(1-1) Therefore, an object of the present invention is to effectively magnetize all the magnetic bands of the face plate, thereby improving the magnetic attraction force.

(Structure) The present invention provides a face plate having a plurality of magnetic bands extending parallel to each other with a non-magnetic band in between, a bottom plate made of a magnetic material arranged along the face plate, and the bottom plate and the face plate. jr, ■, and :jS2, which are arranged at intersections If corresponding to the magnetic bands, one of the magnetic yokes connects the upper magnetic part and the do side with the non-magnetic part in between. a magnetic yoke magnetically divided into magnetic parts; It is disposed between each magnetic yoke and is movable vertically between the bottom plate and the face plate in a state in which the magnetic pole surface is in contact with the corresponding magnetic yoke with a sliding function, and the magnetization direction is in the arrangement direction of the magnetic yokes. It includes a plurality of permanent magnets arranged in an alternately reversed manner, and an operation mechanism that integrally moves the permanent magnets in the vertical direction. It is characterized in that the face plate (+iii) is switched between the upper plate state and the non-excited state only by the permanent magnet, which is movable in the vertical direction between the bottom plate and the face plate without using it.

(Function and Effect) According to the present invention, the permanent magnet places the magnetic region A; 1iii in the lower position in contact with the magnetic part Since the +"N magnetic band is in a de-energized state, the face plate can be switched between an energized state and a de-energized state by moving the magnet up and down between the face plate and the bottom plate, and In this state, the magnetic flux of the permanent magnet is induced in all of the magnetic bands through the yoke corresponding to the magnetic band. The magnetic flux of the 1111 magnet can be effectively applied to the magnet, thereby increasing the magnetic attraction force.

Further, according to the present invention, the face plate can be switched between the excitation state and the J1 excitation state using only the nT dynamic permanent magnet without using a combination of fixed and movable permanent magnets as in the conventional case. Simplification can be achieved.

(Embodiments) The features of the present invention will become clearer from the following description of the illustrated embodiments.

As shown in FIG. 1, the permanent magnet chuck IO according to the present invention includes a face plate 16 including a non-magnetic member 12 and magnetic pole members 14 made of a magnetic material alternately arranged with the non-magnetic member; It includes a bottom plate 18 made of a resistant material arranged parallel to the , and a number of magnets 20 .

11 Magnetic + 11 Member 12 is face plate t 6 ノ,, 1rIn property I 1? In the illustrated example, the FE pole member 14 defines a magnetic zone 14a of the surface plate 16. In order to subdivide the magnetic zone 14a, the non-magnetic member 12 has a U-shaped cross section as a whole. A non-magnetic member having a planar shape is used, and an auxiliary magnetic band 22a is defined by an auxiliary magnetic pole member 22 received in the non-magnetic member 12. 22 are integrally connected to form i16. By using a medium root-shaped nonmagnetic plate as the nonmagnetic member 12, the auxiliary magnetic pole member 22 can be made unnecessary.

Between the face plate 16 and the bottom plate 18, there is a magnetic pole member 1 that corresponds to the magnetic zone 14a, that is, defines the jA Ia magnetic zone. Corresponding to 4, the first and second magnetic yokes 24 and 26 are arranged in parallel with an interval of lf in an intersecting manner. Each of the magnetic yokes 24 and 26 has its upper end surface in contact with the corresponding magnetic pole member 14 and its lower end surface in contact with the bottom plate 18, so that the magnetic yokes 24 and 26 are magnetically attached to each of the ll11 pole member 14 and the bottom plate 18, respectively. It is combined with Shiraku and has one unit. Each magnetic f1 yoke 24 and 26 has a 5"-shaped block shape, H, L4. The second magnetic yoke 26 is made of a single magnetic member that has a rectangular block shape. 4. The first magnetic yoke 24 has the same circular shape and dimensions as the first magnetic yoke 26, but has approximately the same diameter as the first magnetic yoke 26. The magnetic J41 member 1 is located north of the non-magnetic part with the throat magnetic part 24a provided at the 1t5 position in between. 4, and a tonneau J/A portion located on the F side of the non-magnetic portion and abutting the bottom &18.

1 between each adjacent magnetic yoke 24 and 26.

A recess 32 for the permanent magnet 20 is defined between the adjacent magnetic yokes 24 and 26 by spacers 28 and 30 in which an upper spacer 28 and a lower spacer 30 made of a non-magnetic material are arranged. 1 to 1 in each recess 32. II: A generally rectangular frame 34 made of m-type material connects the upper spacer 28 and the lower spacer 30. It is housed so that it can move up and down 1ζ1 with nil. 1 for each frame 34.

Frame body 34 along the extension direction of magnetic zone 14a of face plate 16 Four rectangular openings 36 (only some of which are shown in the figure) are formed to extend through the frame 34 in the thickness direction thereof, and are aligned in the long direction F of the frame 34 .

1tii The Mizuhisa magnet 20 has a rectangular flat plate shape corresponding to the opening 36, and is magnetized in the thickness direction thereof.

As the magnet 20, a low coercive force m such as an alnico magnet is used. A stone can be used, but in order to apply a magnet with a small plate thickness, it is desirable to use a coercive force magnet such as a ferrite magnet or a rare earth magnet. It is preferable to use a crystalline magnet with a fixed magnetic flux density, such as a similar type magnet.

The permanent magnets 20 are rigidly supported by each frame 34 so as to be aligned in the same magnetization direction, and as shown in FIG. 26 , that is, the magnetic a1 of the face plate 16? The magnetization directions are alternately reversed in the width direction of the region 14a. The magnetic pole faces of each magnet 20 are paired; each yoke 24 and 2 6. The R magnet 20 contacts only the upper magnetic portion 24b of the +a magnetic pole surface that contacts the yoke 24 when in the three-way position (the frame 34 contacts the upper spacer 28). The upper magnetic portion 24b and the lower magnetic portion 24'c are arranged so that when the frame body 34 is in the r-direction position in contact with the do-side spacer 30, the magnetic pole surface that contacts the yoke 24 contacts only the F-direction magnetic portion 24'c. magnetic part 24 The height dimension is less than or equal to the height dimension of c.

Each frame serves as an operating mechanism 38 that physically moves each frame 34 in an upward direction in order to integrally move the magnet 20 supported by the frame 34 in the upward direction. 34 A lifting member 40 that engages with a protrusion 34a formed on one side of the and a crank device 42 for positionably moving the elevating member between its upper position and lower position. The crank device 42 is connected to a drive shaft 46 that can be rotated through an opening 44a formed in one of a pair of side plates 44 made of a non-magnetic material, and is supported by the side root 44 for rotation. In addition, the heavy upper drive gear 46 is provided with a pair of driven gears 48 having an interlock uI function, and each driven gear 48 includes an eccentric roller 50 that is slidably received in a thrower 40a formed on the elevating member 40. By rotating the drive ffI\U46, the elevating member 40 can be raised and lowered via the eccentric roller 50 provided on the driven gear, thereby raising and lowering the frame supporting the magnet 20. the body 34 in the upper position where the frame body abuts the upper spacer 28; The frame body 34 can be integrally moved between a lower position where it abuts the F-side spacer 30 and held at each position.

+iii In the upper position of the frame 34, as shown in FIG. 2, the magnetic pole surface of each permanent magnet 20 that contacts the yoke 24 contacts only the upper magnetic portion 24b. , the magnetic pole member 14 in contact with the yoke 26 and York? Face plate 1 in contact with upper magnetic portion 24b of 4 Each magnetic band 14a of 6 is alternately magnetized to yl, magnetic pole, As a result, the face plate 16 is in an excited state! 5. It is placed. With this excitation, when a magnetic body 52 is placed on the face plate 16", a closed loop magnetic flux as shown in FIG. The magnetic body 52 is attracted to the face plate 16.

In the lower position of the frame 34, as shown in FIG. 3, the magnetic pole surface of each permanent magnet 20 that contacts the yoke 24 contacts only the lower magnetic portion 24c. In this state, a closed loop magnetic flux indicated by reference numeral 56 is formed passing through the yoke 26, the upper magnetic portion 24c of the yoke 24, and the bottom plate 18, whereby the face plate 16 is placed in a de-energized state.

In the chuck 10 according to the present invention, as described above, t+: By moving the i-recorded Mizuhisa magnets 20 in the vertical direction integrally, the face plate 16 is brought into an energized state and a de-energized state! /] 1! ! ! In the excitation state, the magnetic flux of the permanent magnet can be induced into all of the magnetic bands through all the magnetic yokes that define the magnetic bands. The magnetic flux of the magnet can be effectively applied to all the magnetic bands including the outermost magnetic bands located on the right and leftmost sides, thereby increasing the magnetic attraction force.

Furthermore, since the face plate 16 can be switched between the energized state and the de-energized state without using a fixed permanent magnet, This eliminates the need for fixed permanent magnets as in the past. The configuration can be simplified.

Furthermore, when the first face plate 16 is energized, each permanent magnet is located near the face plate, so that leakage magnetic flux in the magnetic flux path from the magnet to the face plate and in the opposite direction can be suppressed to a minimum. Prevents decrease in adsorption force due to magnetic flux. F, the magnetic material can be reliably attracted and held.

In the above description, an example has been described in which each of the five-sided magnetic pole members 14, the upper magnetic portion 24b of the magnetic yoke 24, and the magnetic yoke 26 are individually formed. 4, the L-direction magnetic portion 24b of the magnetic yoke 24, and the magnetic yoke 26 can be formed integrally with each other, thereby increasing the mechanical rigidity of the chuck IO.

Moreover, as the operating mechanism 38, various operating mechanisms for integrally moving the magnet in the upward direction can be applied instead of the above-mentioned example.

[Brief explanation of drawings]

FIG. 1 is a perspective view showing a permanent magnet chuck according to the invention in an exploded state with a part thereof cut away, and FIGS. The figures are longitudinal sectional views showing the chuck shown in FIG. 1 in an energized state and in a non-energized state, respectively. 10: Permanent magnetic chuck, 12a: Non-magnetic zone, 14a :m zone, 16: face plate, 18: bottom plate. 20: Permanent magnet, 24.26: Magnetic yoke. 24a: non-magnetic part, 24b=upper magnetic part, 24C: Lower magnetic part, 34: frame, 38: operating mechanism.